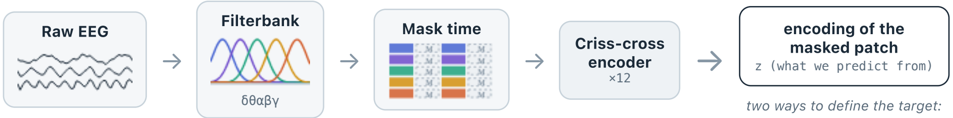
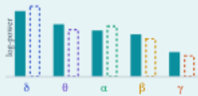
What does the model predict for a masked patch — and why band power, not the band signal?



two ways to define the target:

OURS **Predict masked patch's band power — from context**



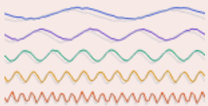
5 numbers / patch · **inferred from context**
(patch is hidden → infer from other time & channels)

target = **fixed filterbank** power (frozen)

many waveforms → **this same summary:** we keep the invariant

✓ To fill in a **hidden** patch's band power the encoder must represent the **spectral state** of the recording (delta-dominated? spindle-rich?) — exactly what sleep / clinical labels are. Phase & fine shape (the part that differs between same-power waveforms) is task-irrelevant and discarded. A **cloze task on band power**: learn the spectral meaning, not the waveform.

ALTERNATIVE **Reconstruct the band signal**



~1280×5 samples / patch
(every point of every band)

target = full band-domain waveforms

✕ This is **reconstruction**: to match the waveform the model must fit **phase, fine spectral shape, and noise** — most of which is task-irrelevant. Capacity is spent modelling noise (the failure mode of raw-signal reconstruction, e.g., CBraMod).

Band-power prediction is only the **pretext task**. After pretraining we **discard the prediction head** and keep the **encoder** — that encoder is the foundation model that transfers to downstream tasks (sleep staging, depression, mental workload). We never reconstruct anything at inference.